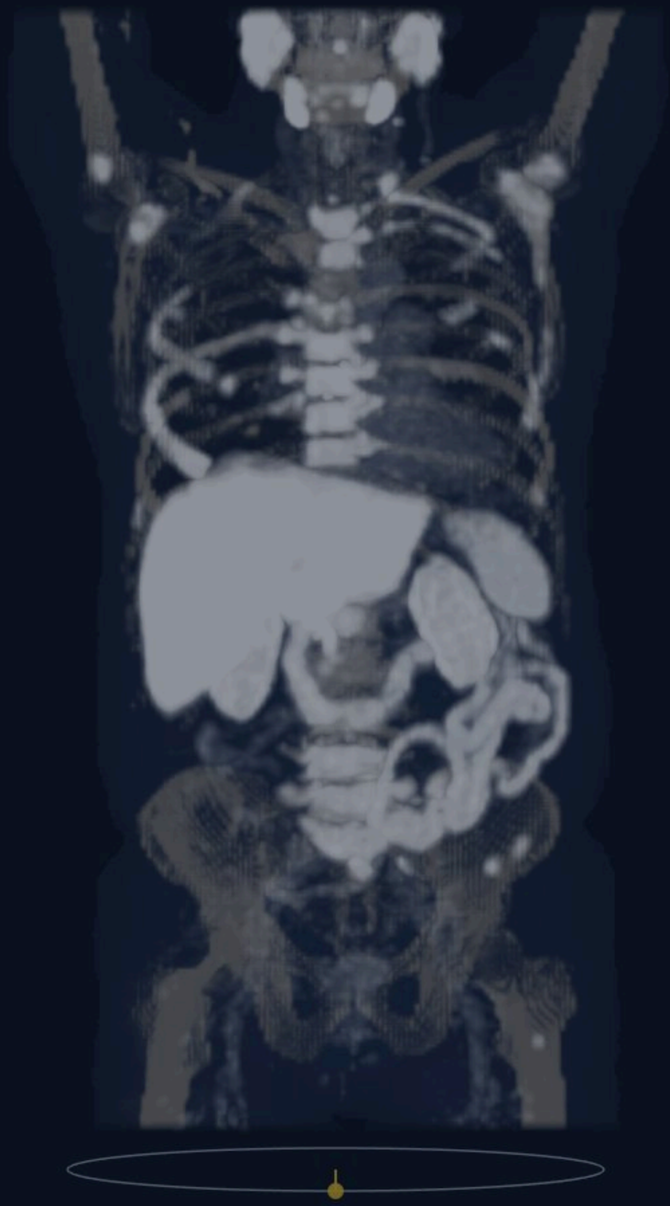
Honours seminar · The University of Western Australia

Segmenting Metastatic Prostate Cancer in Whole-Body PET Scans with a Hybrid CNN-Mamba Network

James Wigfield

Supervised by Dr Jake Kendrick and Dr Mubashar Hassan

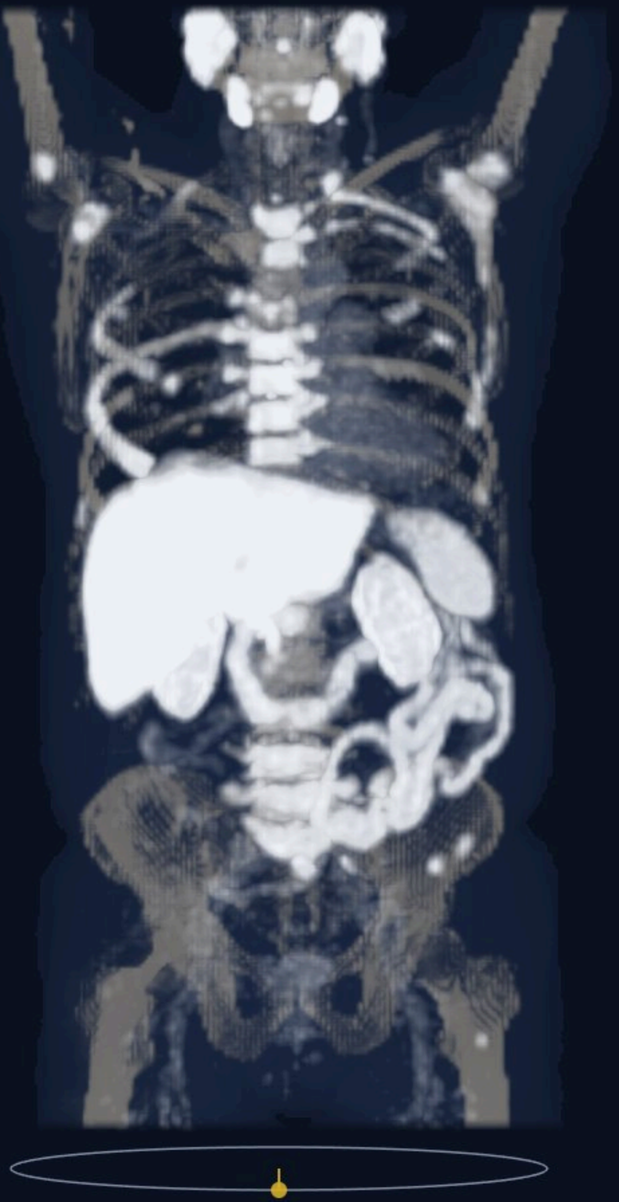


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Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

Why this is hard

This scan holds 52 cancer lesions



Case A · autoPET-III

Outlining every lesion in 3D is called segmentation.

By hand, it is slow.

33%

overlap (Dice) between two experts' outlines of the same 28 scans [1]

CT · bone

PET · tracer uptake

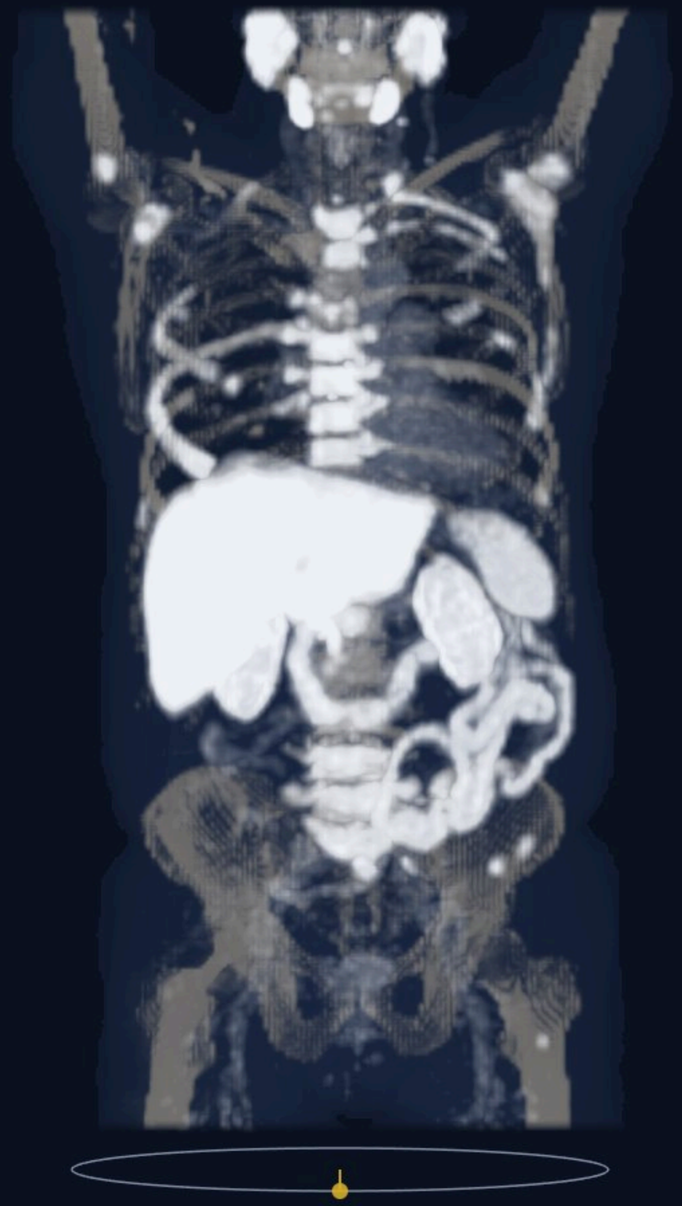
[1] Kendrick et al., Eur J Nucl Med Mol Imaging 2022

Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

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The project in one breath

A hybrid CNN-Mamba network found 82% of lesions at 86% precision



Problem
Lesions are small and scattered.

Idea
More context per window, from Mamba.

Network
Mamba in the deepest three stages, plus an organ task.

Data
697 scans, 130 sealed until the end.

Tests
A fair comparison and a CNN-only twin.

Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

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Previous work · nnU-Net

Today's tools read the body one window at a time

nnU-Net builds a convolutional neural network (CNN). [2]

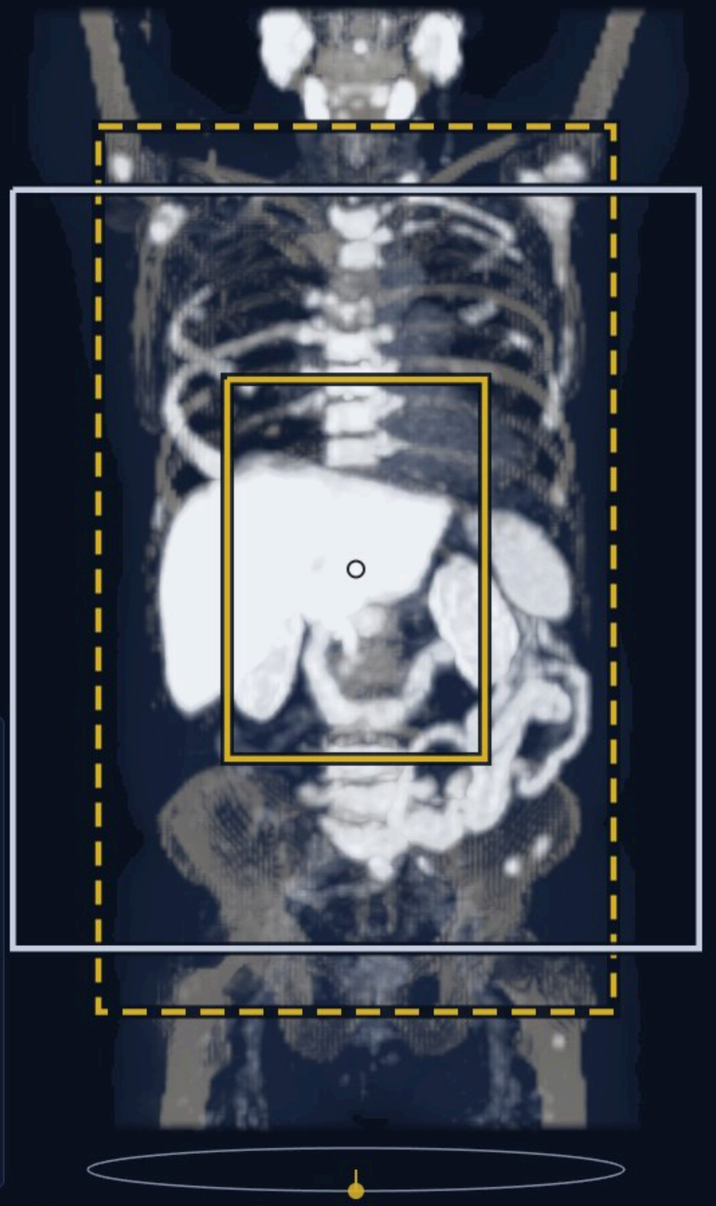
GPU memory caps how big each window can be.

Context only, not comparable

The 2022 nnU-Net cascade (Kendrick et al.) [1]

F1	precision	sensitivity
79.9%	**88.2%**	**73.0%**

on its own 128-scan test set and scoring rule



Windows, to scale

PSMASegmentator
521 mm wide · 576 mm tall

My main window (runs 3 to 5)
196 mm wide · 288 mm tall

My large-window test (run 6)
391 mm wide · 672 mm tall

Coronal view: left-right by head-toe. Depth (front to back) is not shown. The shared centre is illustrative.

[2] Isensee et al., Nature Methods 2021 · [1] Kendrick et al., Eur J Nucl Med Mol Imaging 2022
Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

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The idea, and the question

Mamba reads deep features as one sequence, at linear cost

Research question

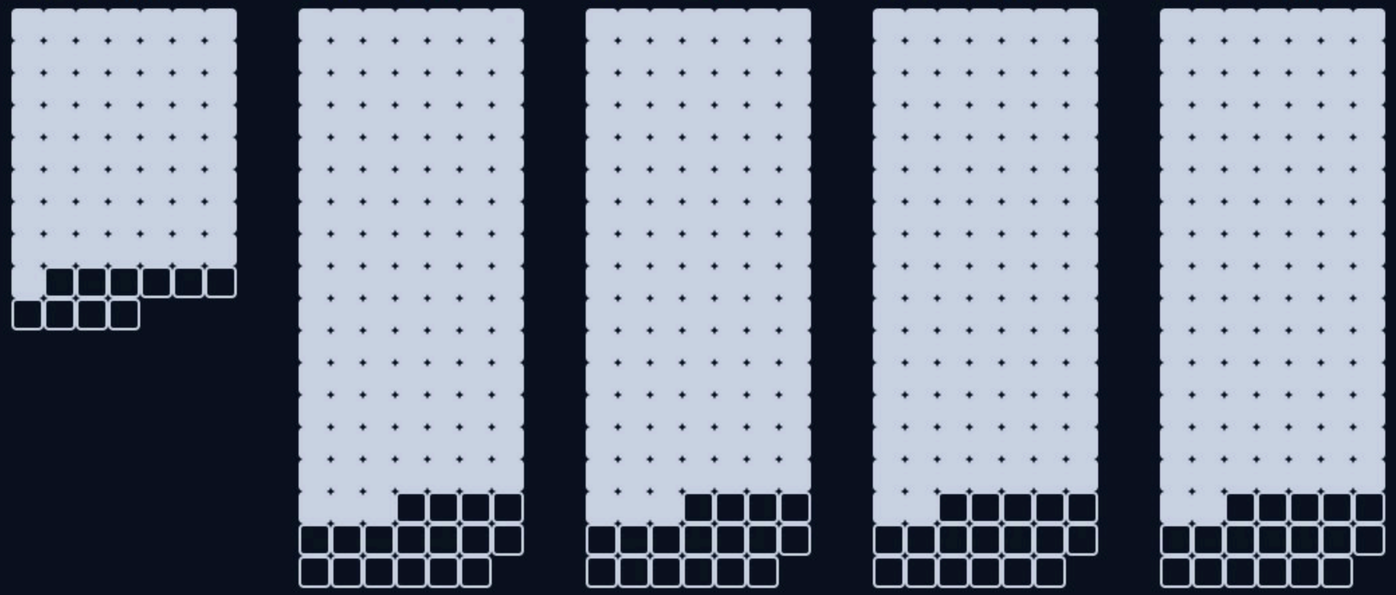
Does a hybrid CNN-Mamba network detect lesions better than PSMASegmentator on scans neither model trained on, and do its Mamba blocks add anything over a CNN-only twin?

H1 · Detection	The hybrid beats PSMASegmentator in pooled F1 on 100 scans neither model trained on.	pending
H2 · Loss	A loss weighted against misses lifts sensitivity and keeps precision high.	pending
H3 · Efficiency	At equal network passes, the hybrid is faster and uses less memory.	pending
H4 · Context	A bigger training window helps the hybrid more than its CNN-only twin.	pending

[3] Gu and Dao, arXiv 2023 · [4] Ma et al., arXiv 2024 · [5] Isensee et al., MICCAI 2024

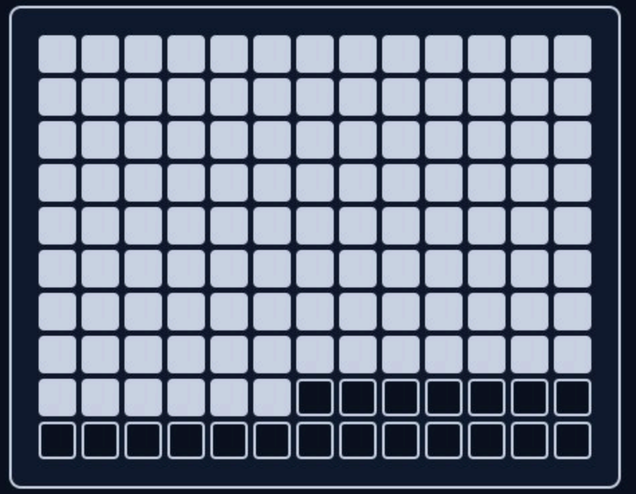
My work · data and protocol

697 scans, 130 locked away until the very end



Grouped by patient, both datasets in every fold. Each model trains on four folds and is checked on the fifth.

Sealed test set · 130 scans



110 autoPET-III + 20 DEEP-PSMA, never trained or tuned on

Pre-registered

Fixed before the test: threshold 0.5, 10-voxel filter, reported rows.

[6] Jeblick et al., Sci Data 2026 · [7] Jackson et al., DEEP-PSMA design document 2025

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My work · the network

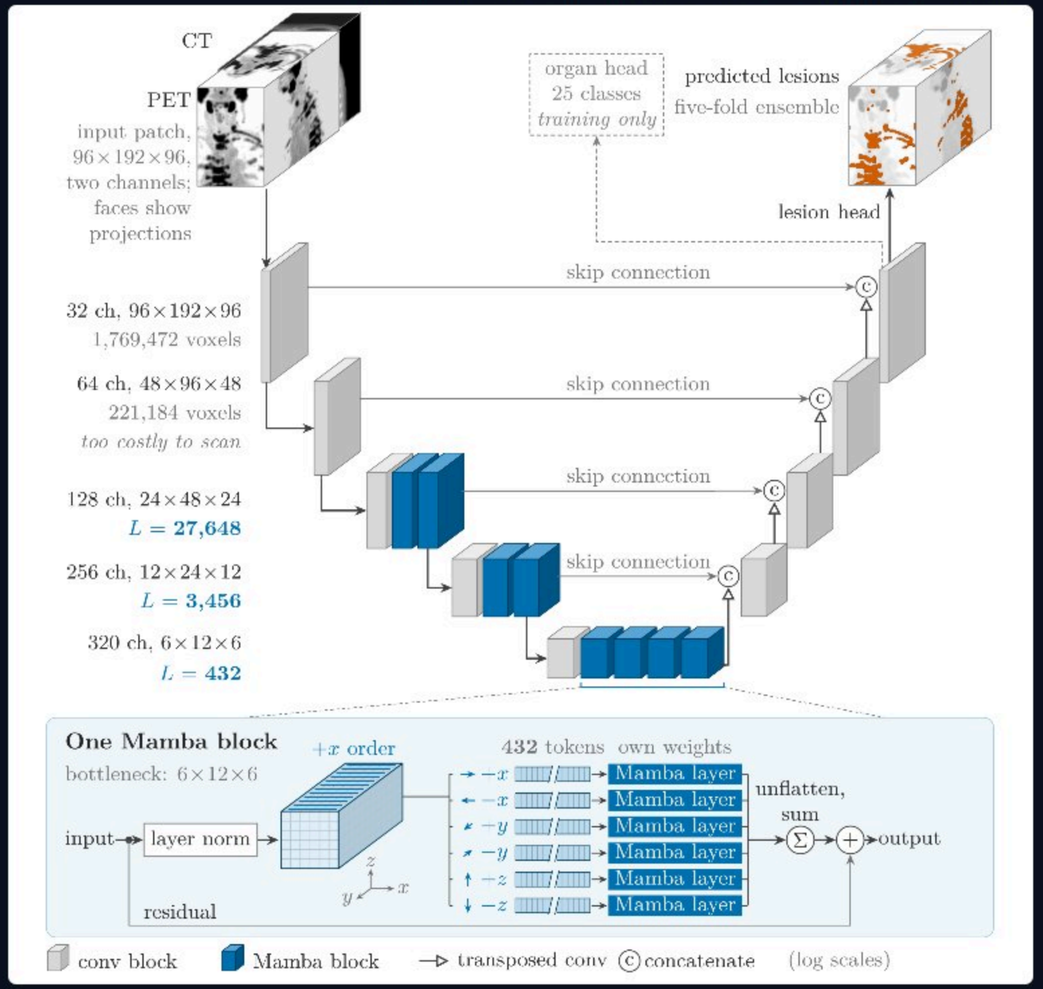
Mamba runs only in the three deepest stages

The whole network

39.7M parameters in five stages.

Convolution at every stage, Mamba in the deepest three, the organ head in training only.

The dissertation's architecture figure, shown as published



[8] Wasserthal et al., Radiol Artif Intell 2023



Case A · autoPET-III

Five models vote, then specks are removed



Average of the 5 models low high

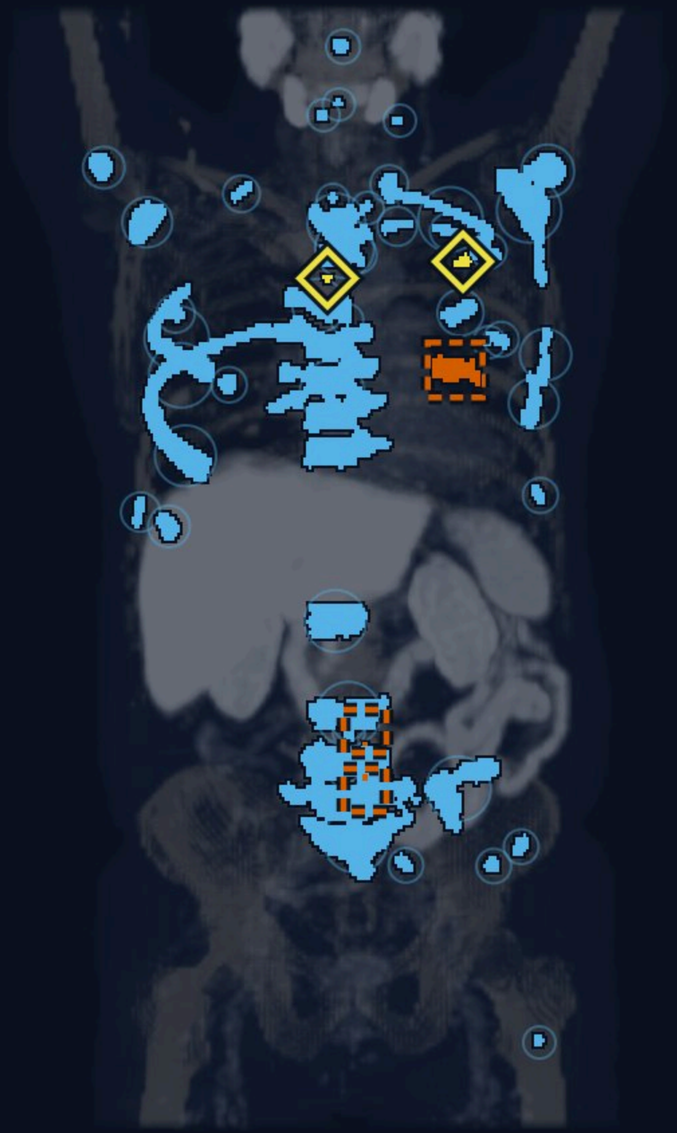
Kept where probability is above 0.5

Regions smaller than 10 voxels (0.12 mL) removed

50 regions → 45

Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

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Case A · autoPET-III

On this scan, the model found nearly every lesion

95.1%	**F1 score** low if either below is low
94.2%	**Sensitivity** found ÷ (found + missed)
96.1%	**Precision (PPV)** found ÷ (found + false alarms)
82.3%	**Dice (DSC)** outline overlap, voxel by voxel; false alarms count against it

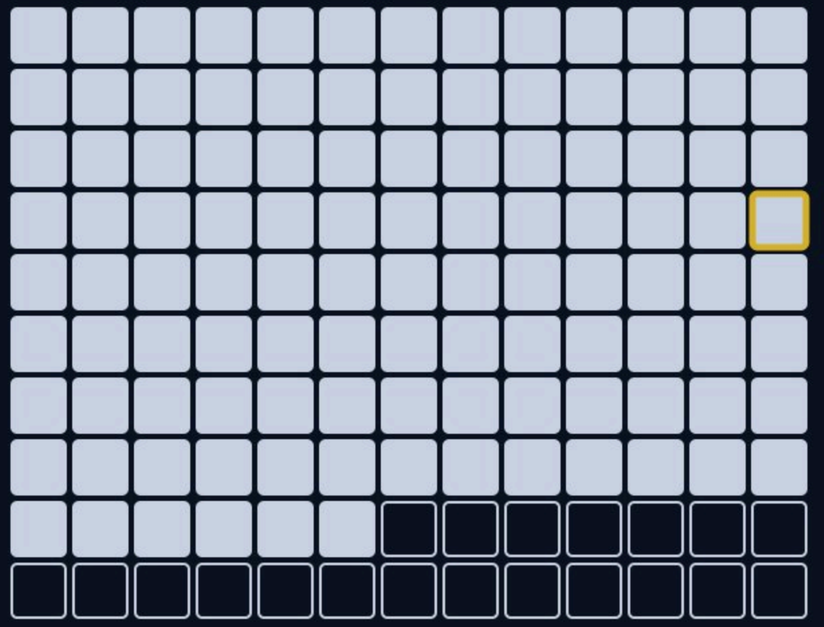
All 130 test scans, lesions pooled: F1 84.3%

Picked from the results table before any image was viewed: the best F1 among autoPET-III test scans with 20 or more lesions.

Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

Results · the sealed test

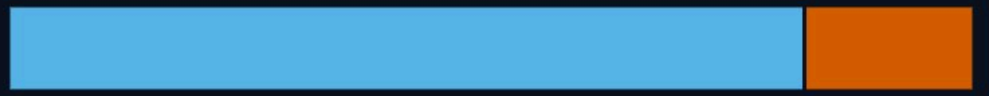
On all 130 sealed scans, the ensemble scored F1 84.3%



autoPET-III DEEP-PSMA

One tile per test scan; positions illustrative.

Every lesion in the test set



sensitivity
82.4%

Lesions found, plus false alarms



precision
86.2%

found missed false alarm

Pooled: lesions added up over all 130 scans.

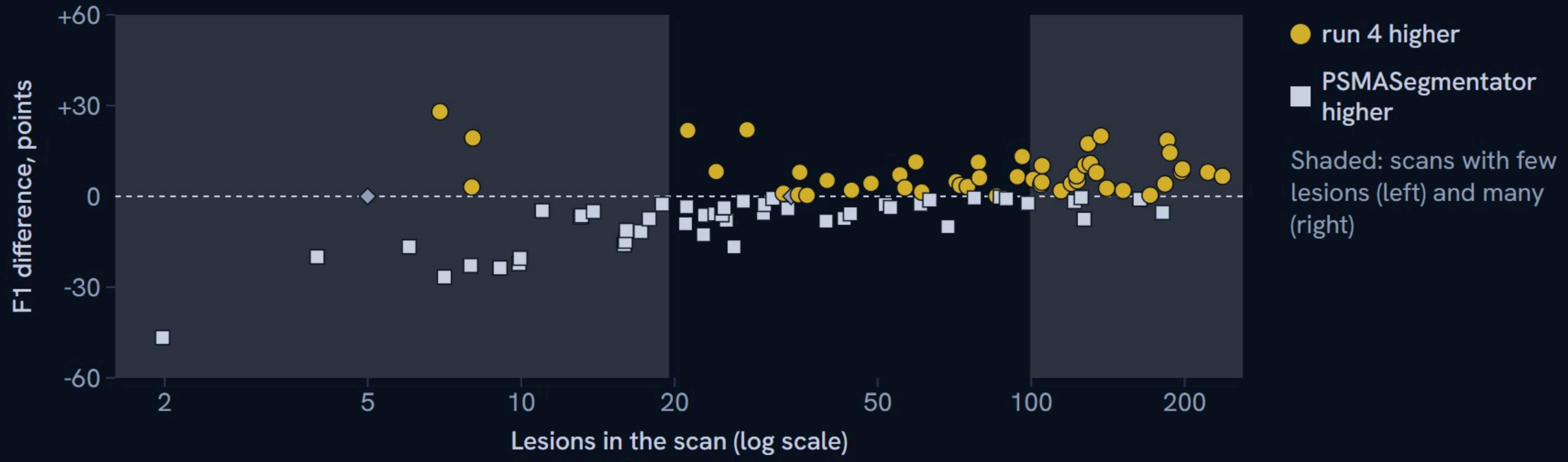
In-domain: the same two datasets the models trained on.

95% interval on F1: 81.4 to 86.5.

Results · the fair comparison

Run 4 found more lesions than PSMASegmentator, and more false alarms

Pooled, run 4 was +3.0 F1 points ahead. Scan by scan, there is no clear difference.



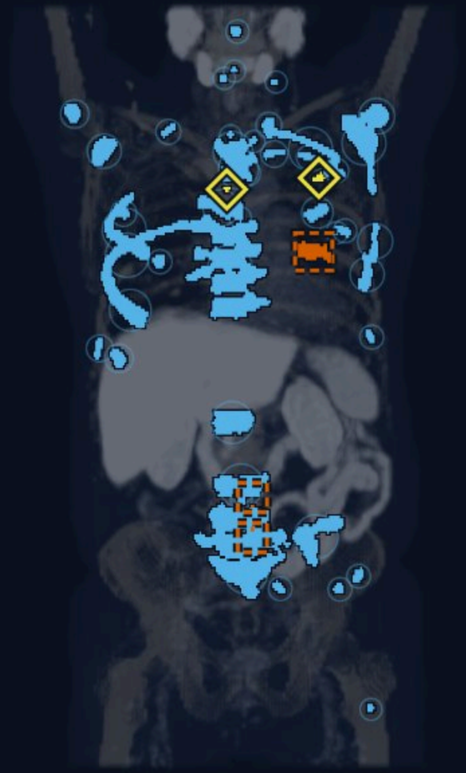
Each dot is one scan. Run 4 better on 48, worse on 50, tied on 2.

Results · the twin ablation

On unseen scans, a CNN-only twin did as well

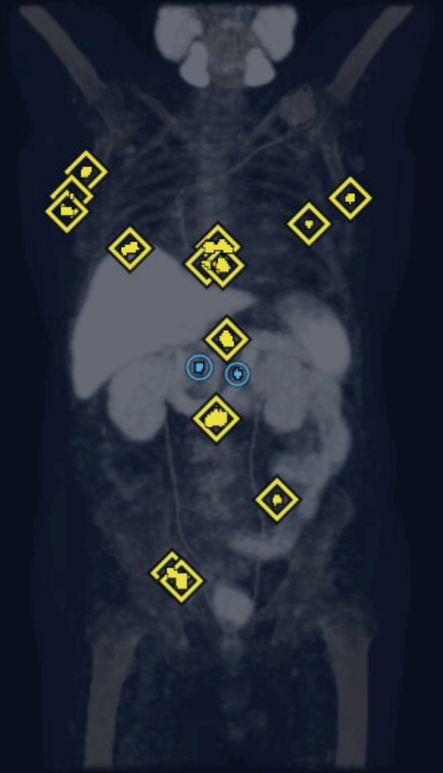
◐	**H1 · Detection**	+3.0 F1 points for run 4 on unseen scans, at lower precision	**supported, as a trade**
✕	**H2 · Loss**	Partial test. Run 2 (older matcher): 71% sensitivity, 61% precision	**not supported**
✕	**H3 · Efficiency**	At equal passes, no faster in mean time and no lower in peak memory	**not supported**
←	**H4 · Context**	On DEEP-PSMA the larger window cost the hybrid more than its twin	**reversed on the primary set**

Case A · autoPET-III



49 of 52 found · 2 false alarms

Case B · autoPET-III



2 of 2 found · 14 false alarms

found · missed · false alarm

Thank you. Questions?

What it means

Run 4 found more lesions than PSMASegmentator on unseen scans, at lower precision; the twin did as well.

Contribution

A fair test on unseen scans; a pre-registered test of what Mamba adds.

Limits

Two public datasets; one run per cell.

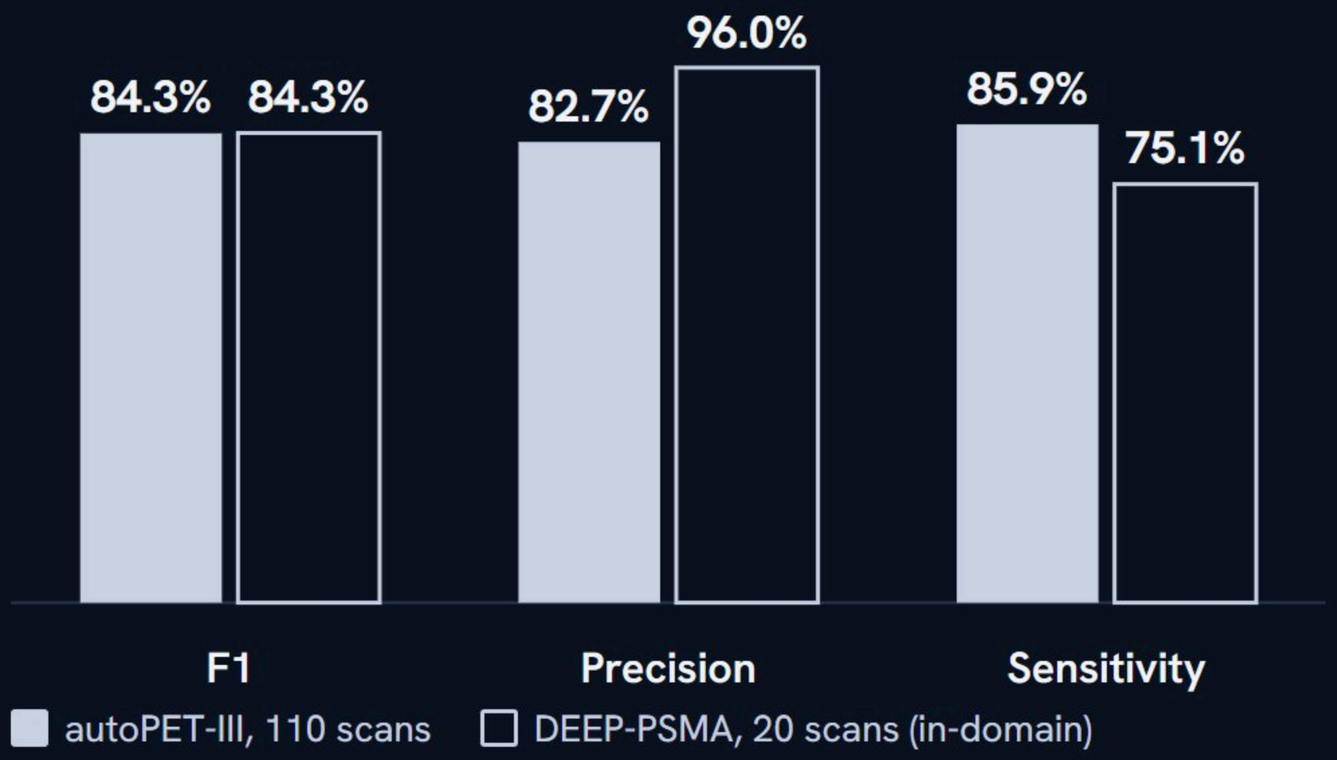
Next

Retrain and test on SCGH.

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Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

Both cohorts reach F1 84.3%, by different routes



- autoPET-III: expert-corrected outlines
- DEEP-PSMA: every voxel at SUV 3 or more, normal uptake painted out
- About 68 pieces per DEEP-PSMA scan

Backup B2

Cross-validation agreed with the sealed test

81.8%

Cross-validation, 567 scans pooled (fold sd 1.5)

84.3%

Sealed test, 130 scans (81.4 to 86.5)

83.1

One fold model alone, plus or minus 0.6

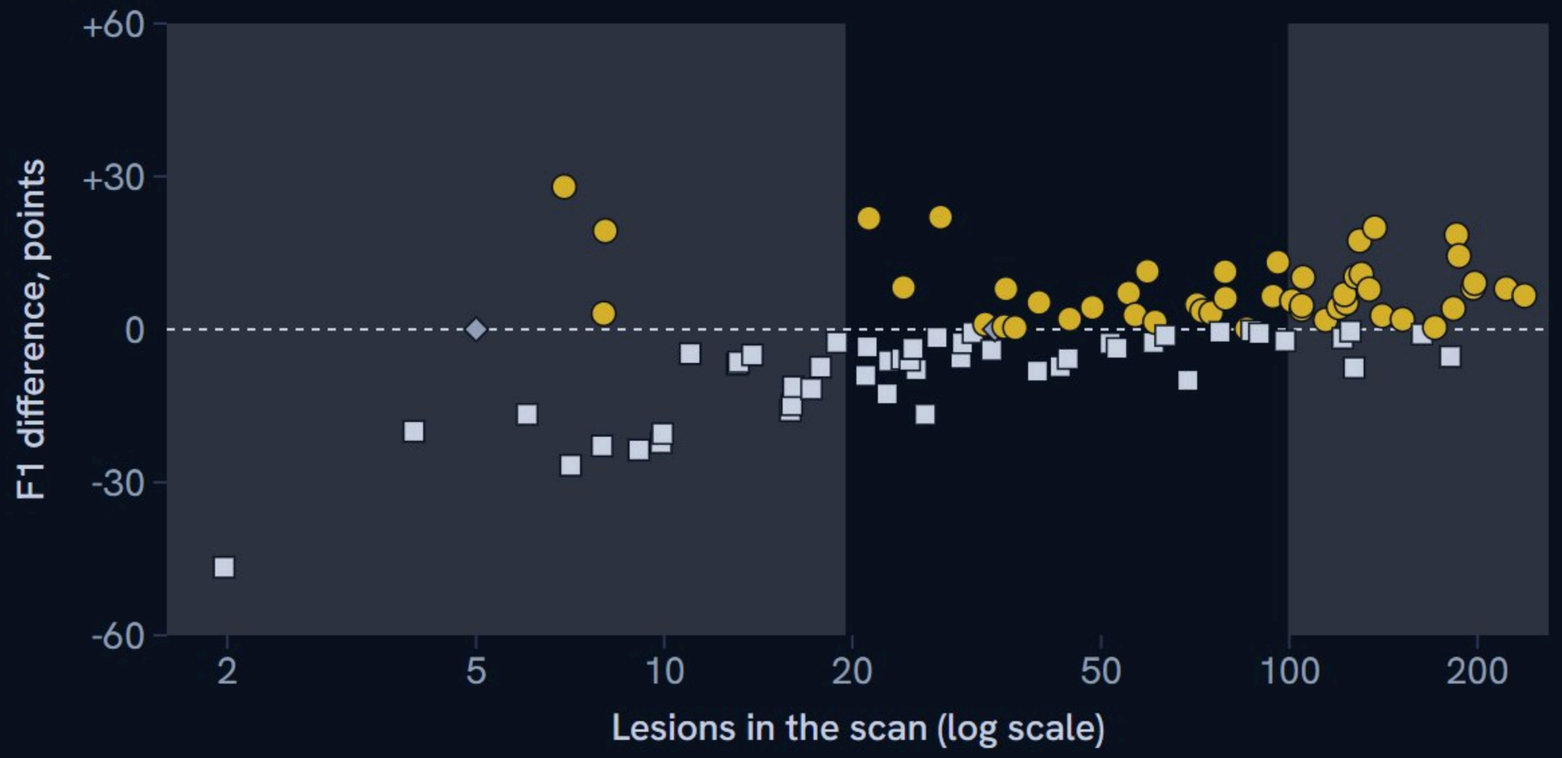
Backup B3

Most recipe differences favour PSMASegmentator

Factor	Run 4 (mine)	PSMASegmentator	Favours
Training scans	430	1,035	PSMASegmentator
Models at inference	one	five averaged	PSMASegmentator
Test-time augmentation	off	eight mirrored passes	PSMASegmentator
Training window	1.77M voxels	4.72M voxels	PSMASegmentator
Parameters	39.7M	102.4M	PSMASegmentator
Optimiser	AdamW	SGD	unclear
Post-processing	none	off for this test	neither
Matching rule, grid, labels	same	same	neither

Backup B4

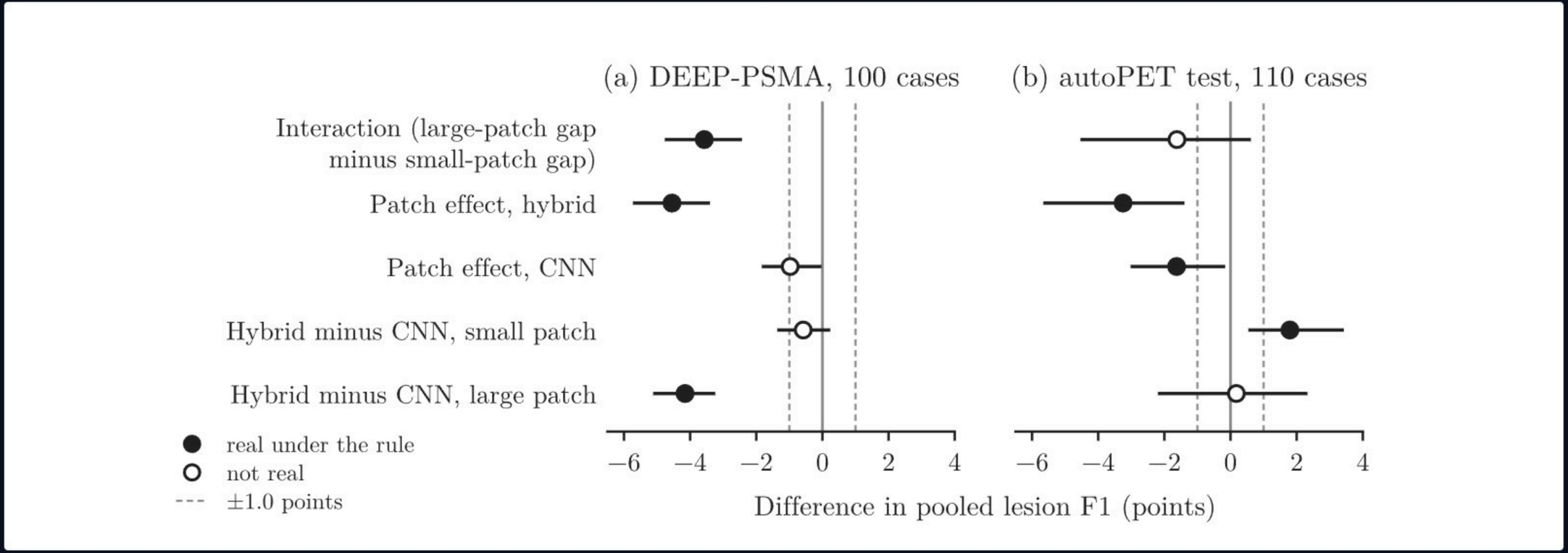
Scan by scan, there is no clear difference



Run 4 minus PSMASegmentator, per scan. Better on 48, worse on 50, tied on 2. Mean -0.6 (-2.8 to 1.6).

Backup B5

Every run 6 comparison, against the pre-registered rule



Real: interval excludes zero and size at least 1.0 point. One run per cell; large window at batch size 1; large hybrid selected at epoch 275.

Backup B6

The speed gap is test-time augmentation, not the architecture

Configuration	Passes per tile	Mean per scan
Hybrid, as scored	5	18.1 s
PSMASegmentator, default	40	43.8 s
PSMASegmentator, augmentation off	5	17.2 s
Hybrid, augmentation on	40	131.2 s

Development runs 1 to 4 fixed stability, resolution and false alarms

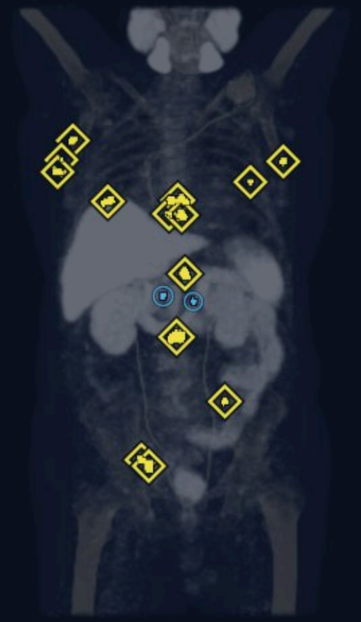
Run	What changed	What it showed
1	recall-weighted loss, no gradient clipping	loss spiked to 4,165; weights later lost
2	clipping, tiled validation (same loss)	precision about 61% (old matcher)
3	2 mm grid, Dice loss, bigger window	false alarms on all seven lesion-free scans
4	organ head	same predictions: F1 73.7 under the old matcher, 85.5 under any-overlap

Backup B8

Mamba sits only where the flattened sequence is short

Stage	Channels	Tokens per direction	Block
1	32	1,769,472	convolution
2	64	221,184	convolution
3	128	27,648	convolution + 2 Mamba
4	256	3,456	convolution + 2 Mamba
5	320	432	convolution + 4 Mamba

Case B: both lesions found, 14 false alarms



Case B · autoPET-III

Against the ground truth

Outlines corrected slice by slice
by an expert, then cross-checked.

2 of 2 lesions found

0 missed

14 false alarms

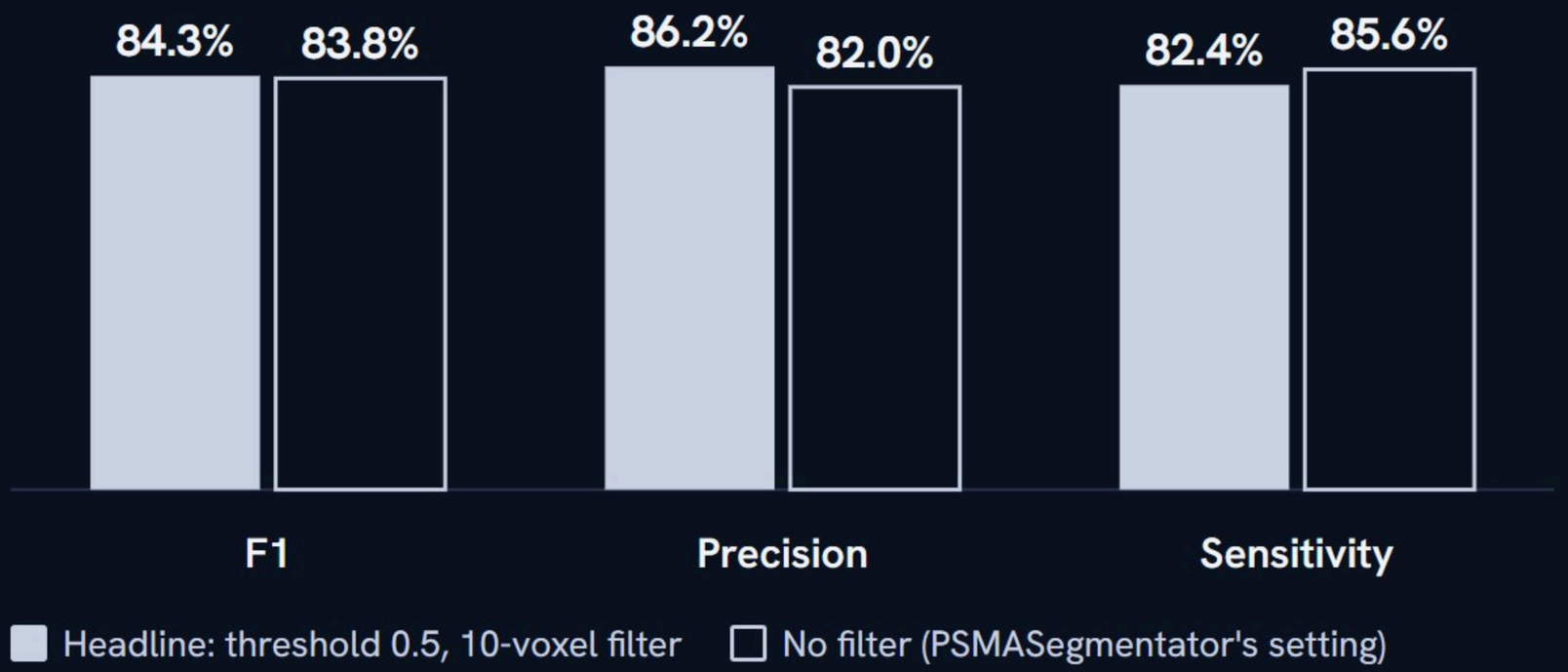
A lesion counts as found if any predicted voxel overlaps it.

- 8 false alarms in bone, 6 in soft tissue
- None in the kidneys or bladder
- Not the worst scan: nine had no lesion found
- Chosen by rule before viewing

Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

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The threshold and size filter were fixed before the test



Backup B11

References

[1] J. Kendrick, R. J. Francis, G. M. Hassan, P. Rowshanfarzad, J. S. L. Ong, and M. A. Ebert, "Fully automatic prognostic biomarker extraction from metastatic prostate lesion segmentations in whole-body [68Ga]Ga-PSMA-11 PET/CT images," European Journal of Nuclear Medicine and Molecular Imaging, vol. 50, pp. 67-79, 2022.

[2] F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, "nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation," Nature Methods, vol. 18, no. 2, pp. 203-211, 2021.

[3] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," arXiv preprint arXiv:2312.00752, 2023.

[4] J. Ma, F. Li, and B. Wang, "U-Mamba: Enhancing long-range dependency for biomedical image segmentation," arXiv preprint arXiv:2401.04722, 2024.

[5] F. Isensee, T. Wald, C. Ulrich, M. Baumgartner, S. Roy, K. Maier-Hein, and P. F. Jaeger, "nnU-Net revisited: A call for rigorous validation in 3D medical image segmentation," in Medical Image Computing and Computer Assisted Intervention (MICCAI 2024), 2024.

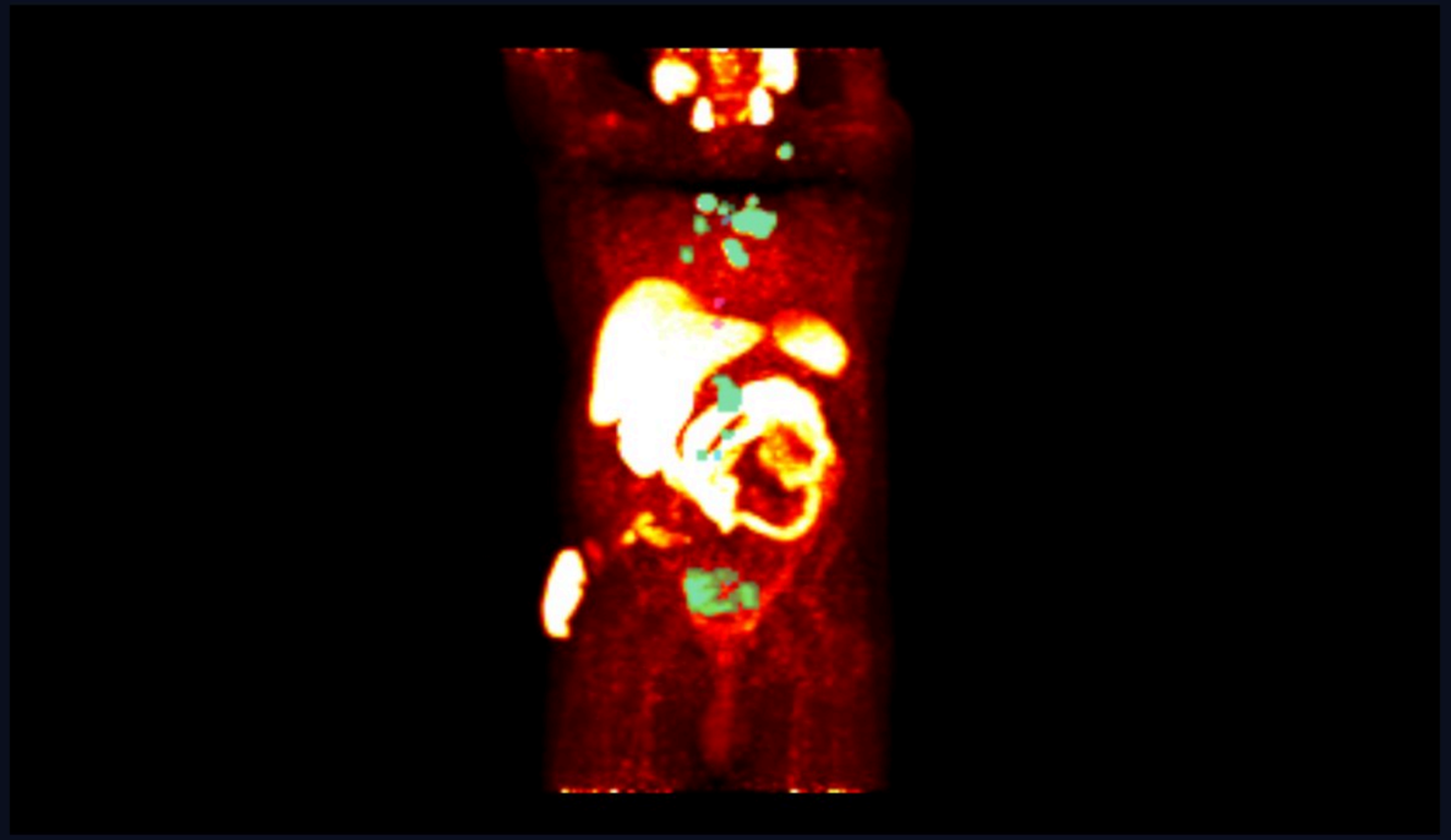
[6] K. Jeblick, B. Schachtner, A. Mittermeier et al., "A whole-body PSMA-PET/CT dataset with manually annotated tumor lesions," Scientific Data, vol. 13, p. 1023, 2026.

[7] P. Jackson, L. McIntosh, J. Buteau, and M. Hofman, "Deep-learning evaluation for enhanced prognostics - prostate specific membrane antigen (DEEP-PSMA)," Challenge design document (Biomedical Image Analysis Challenges initiative), Zenodo, 2025, not peer-reviewed.

[8] J. Wasserthal, H.-C. Breit, M. T. Meyer, M. Pradella, D. Hinck, A. W. Sauter, T. Heye, D. T. Boll, J. Cyriac, S. Yang, M. Bach, and M. Segeroth, "TotalSegmentator: Robust segmentation of 104 anatomic structures in CT images," Radiology: Artificial Intelligence, vol. 5, no. 5, p. e230024, 2023.

Backup B12

The full recorded run in my app



The same run, turned in 3D

- Found
- Missed
- False alarm

The app's own colours. Five-model ensemble, held-out test scan.

Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)